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### Electrode and electrode assembly

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#### Abstract

An electrode includes an active material coating portion coated with an electrode active material on at least one surface of an electrode collector; and an active material non-coating portion which is at one side of the active material coating portion, is not coated with the electrode active material, and includes at least one slit extending from one end to another end thereof, wherein in each slit of the at least one slit, a hole passes through a point at which the respective slit and the active material coating portion meet each other.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/KR2021/001502, filed on Feb. 4, 2021, which claims the benefit of the priority of Korean Patent Application No. 10-2020-0015833, filed on Feb. 10, 2020, both of which are hereby incorporated by reference in their entireties.

### TECHNICAL FIELD

(2) The present invention relates to an electrode and an electrode assembly, and more particularly, to an electrode, which is prevented from being bent or warped after a rolling process is completed, and an electrode assembly.

### BACKGROUND ART

(3) In general, secondary batteries comprise nickel-cadmium batteries, nickel-hydrogen batteries, lithium ion batteries, and lithium ion polymer batteries. Such a secondary battery is being applied to and used in small-sized products such as digital cameras, P-DVDs, MP3Ps, mobile phones, PDAs, portable game devices, power tools, E-bikes, and the like as well as large-sized products requiring high power such as electric vehicles and hybrid vehicles, power storage devices for storing surplus power or renewable energy, and backup power storage devices.

(4) In order to manufacture an electrode assembly, a cathode (hereinafter, referred to as a positive electrode), a separator, and an anode (hereinafter, referred to as a negative electrode) are manufactured and stacked.

(5) Specifically, positive electrode active material slurry is applied to a positive electrode collector, and negative electrode active material slurry is applied to a negative electrode collector to manufacture a positive electrode and a negative electrode. Also, when the separator is interposed and stacked between the manufactured positive electrode and the manufactured negative electrode, unit cells are formed. The unit cells are stacked on each other to form an electrode assembly. Also, when the electrode assembly is accommodated in a specific case, and an electrolyte is injected, the

secondary battery is manufactured.

(6) A process of manufacturing the secondary battery is largely divided into three steps such as an electrode process, an assembly process, and a formation process. In addition, the electrode process is further divided into an active material mixing process, an electrode coating process, a rolling process, a slitting process, and a winding process. Among them, the rolling process is a process of allowing an electrode sheet to pass between two rolling rolls, which are heated at a high temperature, to compress the electrode sheet to a desired temperature and density in order to increase in capacity density by reducing a thickness of the electrode sheet, on which the coating process of an electrode mixture is completed, and increase in adhesion and attachment between an electrode collector and an electrode active material contained in the electrode mixture.

(7) FIG. 1 is a schematic view of an electrode **100** according to the related art, and FIG. 2 is a schematic view illustrating a state in which the electrode **100** is rolled according to the related art.

(8) As illustrated in FIG. 1, the electrode **100** such as a positive electrode **101** (see FIG. 3) and a negative electrode **102** (see FIG. 3) comprise an active material coating portion **1003** coated with an electrode active material **1002** and an active material non-coating portion **1004** that is not coated with the electrode active material **1002**. The active material coating portion **1003** is an area that directly generates electric power and occupies a large area of the electrode **100**. In addition, the active material non-coating portion **1004** is directly cut or is connected to a separate conductive member to form an electrode tab **11**.

(9) According to the related art, when the electrode sheet passes between the two rolling rolls **2** in the rolling process, as illustrated in FIG. 2, the active material coating portion **1003** is relatively thick to receive more heat and pressure, and the active material non-coating portion **1004** is relatively thin to receive less heat and pressure. Therefore, due to the difference in heat and pressure, there is a problem in that the electrode **100** is bent or warped after the rolling process is completed. Particularly, in recent years, as a cell having a high energy density is required, a loading amount of the electrode active material **1002** increases, and this problem has occurred even more.

## DISCLOSURE OF THE INVENTION

### Technical Problem

(10) An object of the present invention for solving the above problem is to provide an electrode, which is capable of being prevented from being bent or warped after a rolling process is completed, and an electrode assembly.

(11) The objects of the present invention are not limited to the aforementioned object, but other objects not described herein will be clearly understood by those skilled in the art from descriptions below.

### Technical Solution

(12) An electrode according to an embodiment of the present invention for solving the above problem comprises: an active material coating portion coated with an electrode active material on at least one surface of an electrode collector; and an active material non-coating portion which is at one side of the active material coating portion, is not coated with the electrode active material, and comprises at least one slit extending from one end to another end thereof, wherein in each slit of the at least one slit, a hole passes through a point at which the respective slit and the active material coating portion meet each other.

(13) Also, each slit of the at least one slit may have a linear shape.

(14) Also, each slit of the at least one slit may have an inclination with respect to a boundary between the active material non-coating portion and the active material coating portion.

(15) Also, each slit of the at least one slit may have a curved shape.

(16) Also, the curved shape may have a constant curvature.

(17) Also, the curved shape may have a plurality of curvature centers.

(18) Also, each slit of the at least one slit may have a wave shape.

(19) Also, a width from one end to another end of each slit of the at least one slit may be changed

regularly or irregularly.

(20) Also, the hole may have at least one of a circular shape, an elliptical shape, or a polygonal shape.

(21) Also, the at least one slit may include a plurality of slits, and one hole may be disposed in one end of each slit of the plurality of slits.

(22) Also, the holes may have shapes different from each other.

(23) In an electrode assembly, in which electrodes and separators are alternately stacked, according to an embodiment of the present invention for solving the above problem, each of the electrodes comprises: an active material coating portion coated with an electrode active material on at least one surface of an electrode collector; and an active material non-coating portion which is at one side of the active material coating portion, is not coated with the electrode active material, and comprises at least one slit extending from one end to another end thereof, wherein in each slit of the at least one slit, a hole passes through a point at which the respective slit and the active material coating portion meet each other.

(24) Also, a slit in a positive electrode of the electrodes and a slit in a negative electrode of the electrodes may have shapes different from each other.

(25) Also, a hole in a positive electrode of the electrodes and a hole in a negative electrode of the electrodes may have shapes different from each other.

(26) Particularities of other embodiments are comprised in the detailed description and drawings.

#### Advantageous Effects

(27) According to the embodiments of the present invention, there are at least the following effects.

(28) Since the slit is formed in the active material non-coating portion of the electrode, the electrode may be prevented from being bent or warped even after the rolling process is completed.

(29) Particularly, the slit having the suitable shape according to the degree of the warpage of the electrode, which is different for the size and kind of the electrode and the applied heat and pressure, may be formed to more efficiently prevent the electrode from being bent or warped.

(30) In addition, since the hole is formed to pass through the point at which the slit and the active material coating portion meet each other, the stress concentrated to one end of the slit may be dispersed to prevent the cracks from occurring toward the active material coating portion.

(31) Particularly, the hole having the suitable shape according to the distribution of the stress, which cause the cracks that are different for the size and kind of the electrode and the applied heat and pressure, may be formed to more efficiently prevent the cracks from occurring toward the active material coating portion.

(32) The effects of the present invention are not limited by the aforementioned description, and thus, more varied effects are involved in this specification.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view of an electrode according to a related art.

(2) FIG. 2 is a schematic view illustrating a state in which the electrode is rolled according to the related art.

(3) FIG. 3 is a schematic view of an electrode assembly according to a first embodiment of the present invention.

(4) FIG. 4 is an assembly view of a pouch type secondary battery according to the first embodiment of the present invention.

(5) FIG. 5 is a schematic view of an electrode according to the first embodiment of the present invention.

(6) FIG. 6 is a schematic view of an electrode according to a second embodiment of the present

invention.

(7) FIG. 7 is a schematic view of an electrode according to a third embodiment of the present invention.

(8) FIG. 8 is a schematic view of an electrode according to a fourth embodiment of the present invention.

(9) FIG. 9 is a schematic view of an electrode according to a fifth embodiment of the present invention.

(10) FIG. 10 is a schematic view of an electrode according to a sixth embodiment of the present invention.

(11) FIG. 11 is a schematic view of an electrode according to a seventh embodiment of the present invention.

(12) FIG. 12 is a schematic view of an electrode according to an eighth embodiment of the present invention.

(13) FIG. 13 is a schematic view of an electrode according to a ninth embodiment of the present invention.

(14) FIG. 14 is a schematic view of an electrode according to a tenth embodiment of the present invention.

#### MODE FOR CARRYING OUT THE INVENTION

(15) Advantages and features of the present invention, and implementation methods thereof will be clarified through following embodiments described with reference to the accompanying drawings. The present invention may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present invention to those skilled in the art. Further, the present invention is only defined by scopes of claims. Like reference numerals refer to like elements throughout.

(16) Unless terms used in the present invention are defined differently, all terms (comprising technical and scientific terms) used herein have the same meaning as generally understood by those skilled in the art. Also, unless defined clearly and apparently in the description, the terms as defined in a commonly used dictionary are not ideally or excessively construed as having formal meaning.

(17) In the following description, the technical terms are used only for explaining a specific exemplary embodiment while not limiting the present invention. In this specification, the terms of a singular form may comprise plural forms unless specifically mentioned. The meaning of “comprises” and/or “comprising” does not exclude other components besides a mentioned component.

(18) Hereinafter, preferred embodiments will be described in detail with reference to the accompanying drawings.

(19) FIG. 3 is a schematic view of an electrode assembly 10 according to a first embodiment of the present invention.

(20) In order to manufacture the electrode assembly 10 according to the first embodiment of the present invention, first, slurry in which a positive electrode active material 1012, a binder, and a plasticizer are mixed is applied to a positive electrode collector 1011 having the form of a metal foil or metal mesh containing aluminum and then is dried and pressed to manufacture a positive electrode 101 (i.e., cathode). In addition, slurry in which a negative electrode active material 1022, a binder, and a plasticizer are mixed is applied to a negative electrode collector 1021 having the form of a metal foil or metal mesh containing copper and then is dried and pressed to manufacture a negative electrode 102 (i.e., anode). The slurry may be usually formed by agitating a granular active material, an auxiliary conductor, a binder, and a plasticizer with a solvent added. At this time, if necessary, the slurry may further comprise a filler. Also, the solvent may be removed in the subsequent process. Each of the positive electrode 101 and the negative electrode 102 may be manufactured in the form of a sheet and then be mounted on a roll.

(21) The electrode assembly **10** is formed by alternately stacking electrodes **100** and separators **103**. Particularly, a stacked structure, in which the separator **103** is interposed between the electrodes **100** or disposed at a left or right side of any one electrode **100** so as to insulate the two types of electrodes **100**, which are manufactured into the positive electrode **101** and the negative electrode **102**, from each other, may be provided. In the stacked structure, the positive electrode **101** and the negative electrode **102**, each of which has a predetermined standard, may be stacked with the separator **103** interposed therebetween or may be wound in the form of a jelly-roll. Alternatively, when the separator **103** is interposed between the manufactured positive electrode **101** and the manufactured negative electrode **102** so as to be stacked, unit cells may be formed. Then, the unit cells may be stacked on each other to form the electrode assembly **10** as illustrated in FIG. 3.

(22) The cathode collector **1011** generally has a thickness of 3  $\mu\text{m}$  to 500  $\mu\text{m}$ . The positive electrode collector **1011** is usually made of a material having high conductivity without causing a chemical change. Such a material may be surface-treated with, for example, stainless steel, aluminum, nickel, titanium, calcined carbon, or aluminum or stainless steel on which carbon, nickel, titanium, silver, or the like is surface-treated on a surface thereof, but is not limited thereto. Also, the positive electrode collector **1011** may form a fine unevenness on a surface of the positive electrode collector **1012** to increase in adhesion of the positive electrode active material **1012**. Also, the positive electrode collector **1011** may have various shapes such as a film, a sheet, a foil, a net, a porous body, a foam, or a non-woven fabric.

(23) In case of a lithium secondary battery, the positive electrode active material **1012** may comprise, for example, a layered compound of lithium cobalt oxide ( $\text{LiCoO}_2$ ), lithium nickel oxide ( $\text{LiNiO}_2$ ), etc., or a substituted compound with one or more transition metals; lithium manganese oxide such as  $\text{Li}_{1+x}\text{Mn}_{2-x}\text{O}_4$  ( $x$  is 0 to 0.33),  $\text{LiMnO}_3$ ,  $\text{LiMn}_2\text{O}_3$ ,  $\text{LiMnO}_2$ , etc.; lithium copper oxide ( $\text{Li}_2\text{CuO}_2$ ); vanadium oxide such as  $\text{LiV}_3\text{O}_8$ ,  $\text{LiFe}_3\text{O}_4$ ,  $\text{V}_2\text{O}_5$ ,  $\text{Cu}_2\text{V}_2\text{O}_7$ , etc.; Ni site-type lithium nickel oxide represented by Chemical Formula of  $\text{LiNi}_{1-x}\text{M}_x\text{O}_2$  ( $\text{M}=\text{Co}, \text{Mn}, \text{Al}, \text{Cu}, \text{Fe}, \text{Mg}, \text{B}$  or  $\text{Ga}$ , and  $x=0.01$  to  $0.3$ ); lithium manganese complex oxide represented by Chemical Formula  $\text{LiMn}_{2-x}\text{M}_x\text{O}_2$  ( $\text{M}=\text{Co}, \text{Ni}, \text{Fe}, \text{Cr}, \text{Zn}$  or  $\text{Ta}$ , and  $x=0.01$  to  $0.1$ ) or  $\text{Li}_2\text{Mn}_3\text{MO}_8$  ( $\text{M}=\text{Fe}, \text{Co}, \text{Ni}, \text{Cu}$  or  $\text{Zn}$ );  $\text{LiMn}_2\text{O}_4$  in which a portion of  $\text{Li}$  is substituted with alkaline earth ions; a disulfide compound;  $\text{Fe}_2(\text{MoO}_4)_3$ , and the like. However, this embodiment is not limited thereto.

(24) The conductive agent is generally added at 1% to about 50% by weight based on the total weight of the mixture comprising the cathode active material **1012**. The conductive agent is usually made of a material having conductivity without causing a chemical change. The conductive agent may comprise, for example, conductive materials such as: graphite such as natural graphite and artificial graphite; carbon black such as acetylene black, Ketjen black, furnace black, channel black, lamp black, and thermal black; conductive fiber such as carbon fiber and metal fiber; metal powder such as carbon fluoride, aluminum, and nickel powder; conductive whisker such as zinc oxide and potassium titanate; conductive oxide such as titanium oxide; or polyphenylene derivatives.

(25) The binder is a component assisting the bonding of the active material to the conductive material and the bonding to the collector and is commonly added at 1 wt % to 50 wt % based on the total weight of the mixture comprising the cathode active material **1012**. Examples of the binder may comprise polyfluoro vinylidene, polyvinyl alcohol, carboxymethyl cellulose (CMC), starch, hydroxypropyl cellulose, regenerated cellulose, polyvinyl pyrrolidone, tetrafluoroethylene, polyethylene, polypropylene, ethylene-propylene-diene polymer (EPDM), sulfonated EPDM, styrene butadiene rubber, fluorine rubber, various copolymers, and the like.

(26) The filler is optionally used as a component that inhibits expansion of the positive electrode **101**. Also, a general filler may be used if the filler is a fibrous material without causing the chemical change. Examples of the filler may comprise olefin polymers such as polyethylene and polypropylene; and fibrous materials such as glass fibers and carbon fibers.

(27) The anode collector **1021** generally has a thickness of 3  $\mu\text{m}$  to 500  $\mu\text{m}$ . The negative electrode collector **1021** is usually made of a material having conductivity without causing a chemical change. Examples of the material comprise copper, stainless steel, aluminum, nickel, titanium, calcined carbon, copper or stainless steel on which carbon, nickel, titanium, silver, or the like is surface-treated on a surface thereof, or aluminum-cadmium alloys. In addition, the negative electrode collector **1021** may form fine unevenness on the surface of the negative electrode collector **1021** to increase in bonding force of the negative electrode active material **1022**. Also, the negative electrode collector **1021** may have various shapes such as a film, a sheet, a foil, a net, a porous body, a foam, or a non-woven fabric.

(28) The negative electrode active material **1022** may comprise, for example, carbon such as non-graphitized carbon, graphite-based carbon, etc.; a metal complex oxide such as  $\text{Li}_{0.5}\text{Fe}_{0.5}\text{O}_{1.5}$  ( $0 \leq x \leq 1$ ),  $\text{Li}_{0.5}\text{WO}_{1.5}$  ( $0 \leq x \leq 1$ ),  $\text{Sn}_{0.5}\text{Me}_{0.5}\text{O}_{1.5}$  ( $\text{Me: Mn, Fe, Pb, Ge; Al, B, P, Si, elements found in Group 1, Group 2 and Group 3 in a periodic table, halogen; } 0 < x \leq 1; 1 \leq y \leq 3; 1 \leq z \leq 8$ ), etc.; a lithium metal; a lithium alloy; a silicon-based alloy; a tin-based alloy; a metal oxide such as  $\text{SnO}$ ,  $\text{SnO}_2$ ,  $\text{PbO}$ ,  $\text{PbO}_2$ ,  $\text{Pb}_2\text{O}_3$ ,  $\text{Pb}_3\text{O}_4$ ,  $\text{Sb}_2\text{O}_3$ ,  $\text{Sb}_2\text{O}_4$ ,  $\text{Sb}_2\text{O}_5$ ,  $\text{GeO}$ ,  $\text{GeO}_2$ ,  $\text{Bi}_2\text{O}_3$ ,  $\text{Bi}_2\text{O}_4$ ,  $\text{Bi}_2\text{O}_5$ , etc.; a conductive polymer such as polyacetylene, etc.; Li—Co—Ni-based material, and the like.

(29) A generally known polyolefin separator or a composite separator in which an organic or inorganic composite layer is formed on an olefin base material may be used as the separator, which insulates the electrodes **100**, between the positive electrode **101** and the negative electrode **102**, but is not particularly limited thereto.

(30) According to a first embodiment of the present invention, the electrode assembly **10** having the above structure is accommodated in a battery case **13**, and then, an electrolyte is injected to manufacture the secondary battery **1**.

(31) FIG. 4 is an assembly view of the pouch type secondary battery **1** according to the first embodiment of the present invention.

(32) As illustrated in FIG. 4, a pouch type secondary battery **1** according to the first embodiment of the present invention comprises an electrode assembly **10**, in which electrodes **100** such as a positive electrode **101** and a negative electrode **102** and a separator **103** are stacked, and a pouch-type battery case **13** accommodating the electrode assembly **10** therein.

(33) In order to manufacture the pouch-type secondary battery **1**, the manufactured electrodes **100** are stacked on both sides of the separator **103** to manufacture the electrode assembly **10** having a predetermined shape. Then, the electrode assembly **10** is inserted into the battery case **13**, and the battery case **13** is sealed after injecting the electrolyte therein.

(34) As illustrated in FIG. 4, the electrode assembly **10** comprises the electrode tabs **11**. The electrode tabs **11** are respectively connected to the positive electrode **101** and the negative electrode **102** of the electrode assembly **10** to protrude to the outside of the electrode assembly **10**, thereby providing a path, through which electrons move, between the inside and outside of the electrode assembly **10**. An electrode collector **1001** of the electrode assembly **10** comprises an active material coating portion **1003** coated with an electrode active material **1002** and an active material non-coating portion **1004** which is not coated with the electrode active material **1002**. Also, an electrode tab **11** may be formed by cutting the non-coating portion or by connecting a separate conductive member to the non-coating portion through ultrasonic welding. As illustrated in FIG. 4, the electrode tabs **11** may protrude from one side of the electrode assembly **10** in the same direction, but the present invention is not limited thereto. For example, the electrode tabs **11** may protrude in directions different from each other.

(35) In the electrode assembly **10**, the electrode lead **12** that supplies electricity to the outside of the secondary battery **1** is connected to the electrode tab **11** through spot welding. Also, a portion of the electrode lead **12** is surrounded by an insulating part **14**. The insulation part **14** may be disposed to



be limited within a sealing part **134**, at which an upper case **131** and a lower case **132** of the battery case **13** are thermally fused, so that the electrode lead **12** is bonded to the battery case **13**. Also, electricity generated from the electrode assembly **10** may be prevented from flowing to the battery case **13** through the electrode lead **12**, and the sealing of the battery case **13** may be maintained. Thus, the insulation part **14** may be made of a nonconductor having non-conductivity, which is not electrically conductive. In general, although an insulation tape which is easily attached to the electrode lead **12** and has a relatively thin thickness is mainly used as the insulation part **14**, the present invention is not limited thereto. For example, various members may be used as the insulation part **14** as long as the members are capable of insulating the electrode lead **12**.

(36) The electrode lead **12** comprises a positive electrode lead **121** having one end connected to a positive electrode tab **111** to extend in a direction in which the positive electrode tab **111** protrudes and a negative electrode lead **122** having one end connected to a negative electrode tab **112** to extend in a direction in which the negative electrode tab **112** protrudes. On the other hand, as illustrated in FIG. 4, all of the other ends of the positive electrode lead **121** and the negative electrode lead **122** protrude to the outside of the battery case **13**. As a result, electricity generated in the electrode assembly **10** may be supplied to the outside. Also, since each of the positive electrode tab **111** and the negative electrode tab **112** is formed to protrude in various directions, each of the positive electrode lead **121** and the negative electrode lead **122** may extend in various directions.

(37) The positive electrode lead **121** and the negative electrode lead **122** may be made of materials different from each other. That is, the positive electrode lead **121** may be made of the same material as the positive electrode collector **1011**, i.e., an aluminum (Al) material, and the negative electrode lead **122** may be made of the same material as the negative electrode collector **1021**, i.e., a copper (Cu) material or a copper material coated with nickel (Ni). Also, a portion of the electrode lead **12**, which protrudes to the outside of the battery case **13**, may be provided as a terminal part and electrically connected to an external terminal.

(38) The battery case **13** is a pouch made of a material having flexibility, which accommodates the electrode assembly **10** therein. Hereinafter, the case in which the battery case **13** is the pouch will be described. When a pouch film **135** having flexibility is drawn by using a punch or the like, a portion of the pouch film **135** is stretched to form a cup part **133** comprising a pocket-shaped accommodation space **1331**, thereby manufacturing the battery case **13**. The battery case **13** accommodates the electrode assembly **10** so that a portion of the electrode lead **12**, i.e., the terminal part is exposed and then is sealed. As illustrated in FIG. 4, the battery case **13** comprises the upper case **131** and the lower case **132**. An accommodation space **1331** in which a cup part **133** is formed to accommodate the electrode assembly **10** may be provided in the lower case **132**, and upper case **131** may cover an upper side of the accommodation space **1331** so that the electrode assembly **10** is not separated to the outside of the battery case **13**. Also, the sealing part **134** is sealed to seal the accommodation space **1331**. Here, the cup part **133** having the accommodation space **1331** may be formed in the upper case **131** to accommodate the electrode assembly **10** in the upper portion. As illustrated in FIG. 4, one side of the upper case **131** and one side of the lower case **132** may be connected to each other. However, the present invention is not limited thereto. For example, the upper case **131** and the lower case **132** may be separately manufactured to be separated from each other.

(39) When an electrode lead **12** is connected to the electrode tab **11** of the electrode assembly **10**, and the insulation part **14** is provided on a portion of the electrode lead **12**, the electrode assembly **10** may be accommodated in the accommodation space **1331** provided in the cup part **133** of the lower case **132**, and the upper case **131** may cover the accommodation space from the upper side. Also, the electrolyte is injected, and the sealing part **134** extending outward from edges of the upper case **131** and the lower case **132** is sealed. The electrolyte may move lithium ions generated by electrochemical reaction of the electrode **100** during charging and discharging of the secondary battery **1**. The electrolyte may comprise a non-aqueous organic electrolyte that is a mixture of a

lithium salt and a high-purity organic solvent or a polymer using a polymer electrolyte. The pouch type secondary battery **1** may be manufactured through the above-described method.

(40) FIG. **5** is a schematic view of an electrode **100a** according to the first embodiment of the present invention.

(41) According to embodiments of the present invention, since a slit **1005a-f** is formed in the active material non-coating portion **1004a-j** of the electrode **100a-j**, the electrode **100a-j** may be prevented from being bent or warped even after a rolling process is completed. Particularly, the slit **1005a-f** having the suitable shape according to a degree of the warpage of the electrode **100a-j**, which is different for a size and kind of the electrode **100a-j** and an applied heat and pressure, may be formed to more efficiently prevent the electrode **100a-j** from being bent or warped. In addition, since a hole is formed to pass through a point at which the slit **1005a-f** and the active material coating portion **1003** meet each other, the stress concentrated to one end of the slit **1005a-f** may be dispersed to prevent cracks from occurring toward the active material coating portion **1003**. Particularly, a hole **1006a-f** having a suitable shape according to distribution of stress, which cause the cracks that are different for the size and kind of the electrode **100a-j** and the applied heat and pressure, may be formed to more efficiently prevent the cracks from occurring toward the active material coating portion **1003**.

(42) For this, the electrode **100a** according to the first embodiment of the present invention comprises: an active material coating portion **1003** coated with an electrode active material **1002** on at least one surface of an electrode collector **1001**; and an active material non-coating portion **1004a** which is formed at one side of the active material coating portion **1003**, is not coated with the electrode active material **1002**, and comprises at least one slit **1005a** extending from one end to the other end thereof. Here, a hole **1006a** is formed to pass through a point at which the slit **1005a** and the active material coating portion **1003** meet each other.

(43) As illustrated in FIG. **5**, the electrode **100a** comprises the active material coating portion **1003**, on which the electrode active material **1002** is applied on at least one surface of the electrode collector **1001**, and the active material non-coating portion **1004a**, on which the electrode active material **1002** is not applied.

(44) The active material coating portion **1003** generates electricity by allowing ions to be injected and discharged through the electrolyte. Also, the active material non-coating portion **1004a**, i.e., the non-coating portion is formed at one side of the active material coating portion **1003**, as described above, and is a distal end on which the electrode active material **1002** is not applied. In addition, an electrode tab **11** may be formed by cutting the non-coating portion or connecting a separate conductive member to the non-coating portion by ultrasonic welding or the like.

(45) According to the first embodiment of the present invention, the active material non-coating portion **1004a** comprises one slit **1005a** extending from one end to the other end thereof. Thus, after the rolling process is completed, even if stretching amounts of the active material coated portion **1003** and the active material non-coating portion **1004a** are different from each other due to a difference in heat and pressure, the slit **1005a** may divide the active material non-coating portion **1004a** into two areas to prevent the electrode **100a** from being bent or warped.

(46) As illustrated in FIG. **5**, one end of the active material non-coating portion **1004a** is adjacent to the active material coating portion **1003**, and the other end of the active material non-coating portion **1004a** faces an opposite side of the active material coating portion **1003**. In addition, the slit **1005a** may extend outward from a boundary between the active material non-coating portion **1004a** and the active material coating portion **1003**. That is, it is preferable that one end of the slit **1005a** is formed at a boundary between the active material non-coating portion **1004a** and the active material coating portion **1003**. If the slit **1005a** does not reach the boundary of the active material coating portion **1003**, an area in which the slit **1005a** is not formed occurs in the active material coating portion **1003**, and this area is because there is a problem of being bent or warped even after the rolling process is completed.

(47) Also, the slit **1005a** is formed in only the active material non-coating portion **1004a**, and it is preferable that the slit **1005a** does not extend to the active material coating portion **1003**. The active material coating portion **1003** receives heat and a pressure having a certain intensity because the area of the active material coating portion **1003** on the electrode **100a** is relatively large, and a thickness of the active material coating portion **1003** in the electrode **100a** is relatively thick. Thus, after the rolling process is completed, the problem that the active material coating portion **1003** is bent or warped does not occur. However, if the slit **1005a** extends up to the active material coating portion **1003**, an effect of the present invention may not be greatly exhibited, and an energy density may also be reduced by reducing the area of the active material coating portion **1003**.

(48) If the width of the slit **1005a** is excessively narrowed, the electrode **100a** may still be bent or warped even after the rolling process. On the other hand, if the width of the slit **1005a** is excessively widened, a cross-sectional area of the electrode tab **11** may decrease, and thus, resistance may increase. Thus, it is preferable to set the width of the slit **1005a**, which is most suitable experimentally according to the size and type of the electrode **100a**, and the temperature and pressure of the electrode **100a** to be rolled.

(49) When the rolling process is performed on the active material coating portion **1003**, the stress may not be concentrated because the other end of the slit **1005a** faces the outside, but the stress may be concentrated at one end of the slit **1005a** to cause cracks toward the active material coating portion **1003**. Thus, according to the first embodiment of the present invention, a hole **1006a** is formed to pass through a point at which the slit **1005a** and the active material coating portion **1003** meets each other. The hole **1006a** may disperse the stress concentrated to one end of the slit **1005a** to prevent the problem that the cracks occur toward the active material coating portion **1003** by the slit **1005a** even after the rolling process is completed.

(50) If the size of the hole **1006a** is also excessively narrowed, the cracks may still occur toward the active material coating portion **1003** after the rolling process. On the other hand, if the size of the hole **1006a** is excessively widened, the energy density may also be reduced by reducing the area of the active material coating portion **1003**.

(51) FIG. **6** is a schematic view of an electrode according to a second embodiment of the present invention.

(52) According to the first embodiment of the present invention, the active material non-coating portion **1004a** comprises one slit **1005a** extending from one end to the other end thereof. In addition, one hole **1006a** is formed at one end of one slit **1005a**.

(53) However, according to a second embodiment of the present invention, as illustrated in FIG. **6**, an active material non-coating portion **1004b** comprises a plurality of slits **1005b** extending from one end to the other end thereof. In addition, one hole **1006b** is formed at one end of each of the plurality of slits **1005b**. Thus, the slits **1005b** divide the active material non-coating portion **1004b** into three (or more in other embodiments) areas to more efficiently prevent an electrode **100b** from being bent or warped.

(54) FIG. **7** is a schematic view of an electrode **100c** according to a third embodiment of the present invention, FIG. **8** is a schematic view of an electrode **100d** according to a fourth embodiment of the present invention, FIG. **9** is a schematic view of an electrode **100e** according to a fifth embodiment of the present invention, and FIG. **10** is a schematic view of an electrode **100f** according to a sixth embodiment of the present invention.

(55) According to the first and second embodiments of the present invention, each of the slits **1005a** and **1005b** has a linear shape. In addition, each of the slits **1005a** and **1005b** may extend outward in a direction perpendicular to a boundary between each of the active material non-coating portions **1004a** and **1004b** and the active material coating portion **1003**. Thus, each of all areas on which the active material non-coating portions **1004a** and **1004b** are divided may have a rectangular shape.

(56) However, a size and type of the electrode **100** may be various, and heat and pressure applied

according to the rolling process may also be various. Thus, it may be inefficient to always form only the slit **1005a**, **b** having the same shape.

(57) According to other embodiments of the present invention, the shape of the slit **1005c-f** may be various. Particularly, according to the third embodiment of the present invention, as illustrated in FIG. 7, a slit **1005c** may have an inclination with respect to a boundary between the active material non-coating portion **1004c** and the active material coating portion **1003**. Thus, the area on which the active material non-coating portion **1004c** is divided may not have a rectangular shape, but may have a trapezoidal shape or a triangular shape.

(58) Alternatively, according to the fourth embodiment of the present invention, as illustrated in FIG. 8, a slit **1005d** may have a curved shape, in particular, may have a curved shape having one curvature center. In this case, the curvature may always be constant, but the curvature may be changed regularly or irregularly. Thus, the slit **1005d** may have various shapes such as an arc shape and an elliptical arc shape.

(59) Alternatively, according to the fifth embodiment of the present invention, as illustrated in FIG. 9, a slit **1005e** may have a curved shape, in particular, may have a curved shape having a plurality of curvature centers. Here, the plurality of curvature centers may be alternately disposed at left and right sides of the slit **1005e** at regular intervals, and thus, the slit **1005e** may have a wave shape. Also, the curves formed for each of the curvature centers may all have the same curvature.

Alternatively, the curves may have different curvatures. That is, the plurality of curvature centers are not limited and may be variously disposed, and the curvature of the curve formed for each curvature center may be formed variously.

(60) Alternatively, according to the sixth embodiment of the present invention, as illustrated in FIG. 10, a width of the slit **1005f** may be changed regularly or irregularly. For example, the width may be uniformly widened from one end to the other end of the slit **1005f**, and conversely, the width may be uniformly narrowed. Alternatively, a phenomenon in which the width is widened and narrowed at regular intervals may be repeated. That is, the width of the slit **1005f** is not limited and may be formed variously.

(61) As described above, a degree of warpage of the electrode **100a-j** is different for a size and kind of the electrode **100a-j** and an applied heat and pressure, and thus, the slit **1005a-f** having a suitable shape according to the degree of the warpage of the electrode **100a-j** may be formed to more efficiently prevent the electrode **100a-j** from being bent or warped.

(62) Also, a plurality of slits **1005a-f** having various shapes may be formed in the active material non-coating portion **1004a-j**. In addition, one hole **1006a-f** may be formed at one end of each of the plurality of slits **1005a-f**.

(63) Furthermore, even in one electrode assembly **10**, the positive electrode **101** and the negative electrode **102** are different in size and material from each other. Thus, the slit **1005a-f** formed in the positive electrode **101** and the slit **1005a-f** formed in the negative electrode **102** may have different shapes from each other.

(64) FIG. 11 is a schematic view of an electrode **100g** according to a seventh embodiment of the present invention, FIG. 12 is a schematic view of an electrode **100h** according to an eighth embodiment of the present invention, FIG. 13 is a schematic view of an electrode **100i** according to a ninth embodiment of the present invention, and FIG. 14 is a schematic view of an electrode **100j** according to a tenth embodiment of the present invention.

(65) According to the first to sixth embodiments of the present invention, the hole **1006a** has a circular shape. However, a size and type of the electrode **100a-j** may be various, and heat and pressure applied according to the rolling process may also be various. Thus, it may be inefficient to always form only the hole **1006a** having the circular shape.

(66) For example, if the hole **1006a** has the circular shape, stress may be uniformly dispersed, but durability of peripheral areas of the hole **1006a-f** may be different according to the types of electrode **100a-j**. That is, there may be an area in which the durability is strong, and thus, cracks are

not easily generated even under large stress, whereas there may be an area in which the durability is weak, and thus, cracks are easily generated even under small stress.

(67) According to other embodiments of the present invention, the shape of the hole **1006a-f** may be various. For example, the hole **1006a-f** may have at least one shape of a circular, elliptical, or polygonal shape. Particularly, each of the holes **1006c** and **1006d** may have an elliptical shape, and according to the seventh embodiment of the present invention, as illustrated in FIG. **11**, the hole **1006c** has a long axis having an elliptical shape, which is formed parallel to the slit **1005a**. Alternatively, according to the eighth embodiment of the present invention, as illustrated in FIG. **12**, the hole **1006d** may have a short axis having the elliptical shape, which is formed parallel to the slit **1005a**.

(68) In the elliptical-shaped holes **1006c** and **1006d**, the stress is widely dispersed to a corner having a small curvature, and the stress is narrowly dispersed to a corner having a large curvature. Thus, each of the holes **1006c** and **1006d** may be formed so that the corner having the small curvature faces an area having relatively strong durability, and the corner having the large curvature faces an area having relatively weak durability.

(69) Alternatively, the hole **1006e-f** may have a polygonal shape, and according to the ninth embodiment of the present invention, as illustrated in FIG. **13**, the hole **1006e** may be formed in a pentagonal shape. Alternatively, according to the tenth embodiment of the present invention, as illustrated in FIG. **14**, the hole **1006f** may be formed in a triangular shape.

(70) In the polygonal-shaped holes **1006e** and **1006f**, the stress is widely dispersed to a corner of a straight line, and the stress is narrowly dispersed to a vertex. Thus, each of the holes **1006c** and **1006d** may be formed so that the vertex faces an area having relatively strong durability, and the corner of the straight line faces an area having relatively weak durability.

(71) As described above, distribution of the stress that causes the cracks may be different for the size and kind of the electrode **100a-j** and the applied heat and pressure, and thus, the hole **1006a-f** having a suitable shape according to the distribution of the stress may be formed to more efficiently prevent the cracks from occurring toward the active material coating portion **1003**.

(72) In addition, a plurality of slits **1005a-f** may be formed in the active material non-coating portion **1004**. Here, one hole **1006a-f** having various shapes may be formed at one end of each of the plurality of slits **1005a-f**, and the holes **1006a-f** may have shapes different from each other.

(73) Furthermore, even in one electrode assembly **10**, the positive electrode **101** and the negative electrode **102** are different in size and material from each other. Thus, the hole **1006a-f** formed in the positive electrode **101** and the hole **1006a-f** formed in the negative electrode **102** may have different shapes from each other.

(74) Those with ordinary skill in the technical field of the present invention pertains will be understood that the present invention can be carried out in other specific forms without changing the technical idea or essential features. Therefore, the above-disclosed embodiments are to be considered illustrative and not restrictive. Accordingly, the scope of the present invention is defined by the appended claims rather than the foregoing description and the exemplary embodiments described therein. Various modifications made within the meaning of an equivalent of the claims of the invention and within the claims are to be regarded to be in the scope of the present invention.

(75) TABLE-US-00001 [Description of the Symbols]     1: Secondary battery    10: Electrode assembly    11: Electrode tab    12: Electrode lead    13: Battery case    14: Insulation part 100a-j: Electrode    101: Positive electrode    102: Negative electrode    103: Separator    111: Positive electrode tab    112: Negative electrode tab    121: Positive electrode lead    122: Negative electrode lead    131: Upper case    132: Lower case    133: Cup part    134: Sealing part 1001: Electrode collector    1002: Electrode active material    1003: Active material coating portion 1004a-j: Active material non-coating portion 1005a-f: Slit 1006a-f: Hole 1011: Positive electrode collector    1012: Positive electrode active material    1021: Positive electrode collector    1022: Positive electrode active material    1331: Accommodation space

## Claims

1. An electrode comprising: an active material coating portion coated with an electrode active material on at least one surface of an electrode collector; and an active material non-coating portion which is at one side of the active material coating portion, is not coated with the electrode active material, and comprises at least one slit extending from one end to another end thereof, wherein in each slit of the at least one slit, a hole passes through a point at which the respective slit, the active material non-coating portion, and the active material coating portion meet each other.
  2. The electrode of claim 1, wherein each slit of the at least one slit has a linear shape.
  3. The electrode of claim 2, wherein each slit of the at least one slit has an inclination with respect to a boundary between the active material non-coating portion and the active material coating portion.
  4. The electrode of claim 1, wherein each slit of the at least one slit has a curved shape.
  5. The electrode of claim 4, wherein the curved shape has a constant curvature.
  6. The electrode of claim 4, wherein the curved shape has a plurality of curvature centers.
  7. The electrode of claim 6, wherein each slit of the at least one slit has a wave shape.
  8. The electrode of claim 1, wherein a width from one end to another end of each slit of the at least one slit is changed regularly.
  9. The electrode of claim 1, wherein the hole has at least one of a circular shape, an elliptical shape, or a polygonal shape.
  10. The electrode of claim 1, wherein the at least one slit includes a plurality of slits, and one hole is disposed in one end of each slit of the plurality of slits.
  11. The electrode of claim 10, wherein the holes have shapes different from each other.
  12. An electrode assembly, in which electrodes and separators are alternately stacked, wherein each of the electrodes is an electrode according to claim 1.
  13. The electrode assembly of claim 12, wherein a slit in a positive electrode of the electrodes and a slit in a negative electrode of the electrodes have shapes different from each other.
  14. The electrode assembly of claim 12, wherein a hole in a positive electrode of the electrodes and a hole in a negative electrode of the electrodes have shapes different from each other.
  15. The electrode of claim 1, wherein a width from one end to another end of each slit of the at least one slit is changed irregularly.
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